Electroencephalogram-Based Machine Learning Approaches for Detecting Major Depressive Disorder: A Review (2021–2025)

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***Abstract*—Major Depressive Disorder (MDD) is a chronic mental health condition that affects a significant portion of the population and is especially common among young adults. Accurate diagnosis remains challenging due to the subjective nature of traditional assessment methods. Electroencephalography has emerged as a promising non-invasive technique for analyzing brain activity related to depression. Recent studies have applied machine learning and deep learning techniques to electroencephalography data to support computer aided diagnosis of Major Depressive Disorder. This study reviews electroencephalography-based machine learning research published between 2021 and 2025, highlighting methodological trends and key challenges. While advanced models demonstrate promising performance, limitations such as small datasets, signal noise, and limited interpretability remain. Future research should focus on developing larger datasets and clinically transparent models to support real world application.**

***Keywords*—Major Depressive Disorder (MDD), Machine learning, Electroencephalogram (EEG), Computer-aided diagnosis (CAD)**

I. Introduction

Machine Learning (ML) is a computation methods that able to automatically identify patterns within data [1]. ML learn the pattern from the data and could perform classification or prediction tasks without the need to explicitly depend on rule-based programming [2], [3]. Deep Learning (DL) is a subset of ML that used layered neural networks like Convolutional Neural Networks, or CNNs to learn more complex features from the input data, thus eliminating the need for manual selection of features [4], [5]. In recent years, ML methods have been increasingly applied to many fields. Within the medical and clinical domain, numerous studies focus on using ML and DL with electroencephalogram (EEG) signals and machine learning techniques for diagnosing Major Depressive Disorder (MDD) [1], [6]. Deep learning models for diagnosing psychological disorders are gaining popularity because of their accuracy and reliability [5]. While traditional machine learning methods dominated the early research on EEG-based MDD diagnosis, deep learning methods have become a recent hotspot for researchers. More recent research shows a clear shift towards deep learning architectures, including CNN and multi-task learning frameworks, which are capable of learning complex representations directly from EEG data and improving classification performance [4], [5], [7]. These developments indicate a growing research trend toward data-driven and automated approaches for MDD detection. MDD is a clinical term for depression, one of the most prevalent and serious mental health disorder in the world [5]. MDD also referred to as a chronic mental illness which affects people's well-being [8]. There are several symptom of MDD, including low mood, ongoing feelings of sadness, loss of interest in doing hobbies, insomnia or hypersomnia, cognitive function decline, and frequent suicidal thoughts or attempts in severe cases [5], [8]. MDD has become a global concern until now, contributing to the significant increase in suicides and disabilities worldwide. It affects 14% of people on average at some point in their lives, and it is most prevalent among young people, especially women and particularly individuals in the 18 to 29 age range [5], [8]. MDD can be clinically diagnosed. The clinical diagnosis of MDD relies heavily on questionnaire-based approaches and mental status assessments [1]. Psychologists currently diagnose depression based on scale-based interview standards [5], [1]. The examination of mental status also used several common standardized scales like Diagnostic and Statistical Manual for depression (DSM-IV), Hamilton Depression Rating Scale (HDRS), 17-item Hamilton rating scale (HAMD-17), Beck Depression Inventory (BDI), MiniMental State Examination (MMSE), and Mini-International Neuropsychiatric Interview (M.I.N.I) [9], [5], [6], [1]. These diagnostic technique are inconsistent since biomarkers are not used, and it mostly depend on the patients' participation and interpretation [5]. Hence, EGG is emerging solution that can be used to diagnosed MDD as it record the brain activity, inexpensive, and non-invasive [1]. It means, EEG use brain the signal as biological indicator and it is a safe method as it can be done without performing procedure that require cutting or inserting instruments into the body, or causing physical injuries. But, using EEG still have the significant challenges. The major limitations are the small dataset, lack of generalisation, and unclear clinical application of EGG-based [1], [6]. Hence, this paper offers a critical review of existing research on the application of EGG and ML in diagnosing MDD based on published journal paper during the period of 2021 to 2025.

The main objective of this study are to:

a) Analyse current research trends in EEG-based machine learning approaches for Major Depressive Disorder detection from 2021 to 2025.

b) Analyse the challenges faced in this research area and the methods proposed to address them.

c) Analyse existing limitations and propose promising directions for future research in EEG-based ML systems for MDD diagnosis.

II. MATERIALS AND METHODS

A. Data Collection

The datasets for EEG-based machine learning in predicting MDD were obtained from Scopus. It is a large research database made by Elsevier which contains myriad of abstracts and citations of literature covers 330 disciplines area totalling up to 2 million documents. Scopus includes only high-quality, peer-reviewed journals, so the articles have gone through strict review processes and meet strong research standards. It also provides access to raw data, which allows researchers to run their own analyses and produce relevant metrics [10].

Figure 1 is the flowchart that illustrates the methodology used for a bibliometric analysis of research on machine learning, EEG, and Major Depressive Disorder (MDD) from 2021 to 2024. To get the relevant papers, the inclusion criteria were defined. Selected paper were required to focus on electroencephalography (EEG)-based research, using Machine Learning (ML) or Deep learning (DL) as the subset of ML and targeting the diagnosis of MDD. Hence, the search keywords in English were used. The search strings is "Machine Learning AND Major Depressive Disorder AND Electroencephalography AND Signal". This resulting to 2,730 initial documents within all fields. Then, a multi-screening process was carried out. There are several filtering stages that the initial documents undergoes. The first stage, the initial 2,730 documents were filtered to specific search field to only title, abstract and keyword. This exclude 2,624 papers resulting to only 106. From 106, documents are filtered based on year to only keep the documents from 2021 to 2025 in the second stage. This exclude 17 documents, resulting to 89 documents. In the third stage, 26 documents were excluded from 89 documents based on source type (journal), documents type (article), and publication stage (final). Each exclude 19, 6, and 1 documents respectively. This resulting to 63 documents. From 63 documents, subject area exclusion are apply to exclude 11 areas that unrelated. Subject area only limit to Computer Science, Engineering, Neuroscience, Medicine, Health Professions, Psychology, Pharmacology, Toxicology and Pharmaceutics. This fourth stage resulting to 45 documents, removing 18 documents from 63. Keyword inclusion has been applied to the keyword of Machine Learning, Electroencephalography, and Major Depressive Disorder but the number remain unchanged. For the bibliometric analysis, only eight representative studies from 45 documents overall were selected for detailed analysis.

B. Data Analysis and Review Strategy

The eight selected studies were analysed by using a bibliometric method and qualitative critical review. The bibliometric analysis was conducted by utilizing the existing

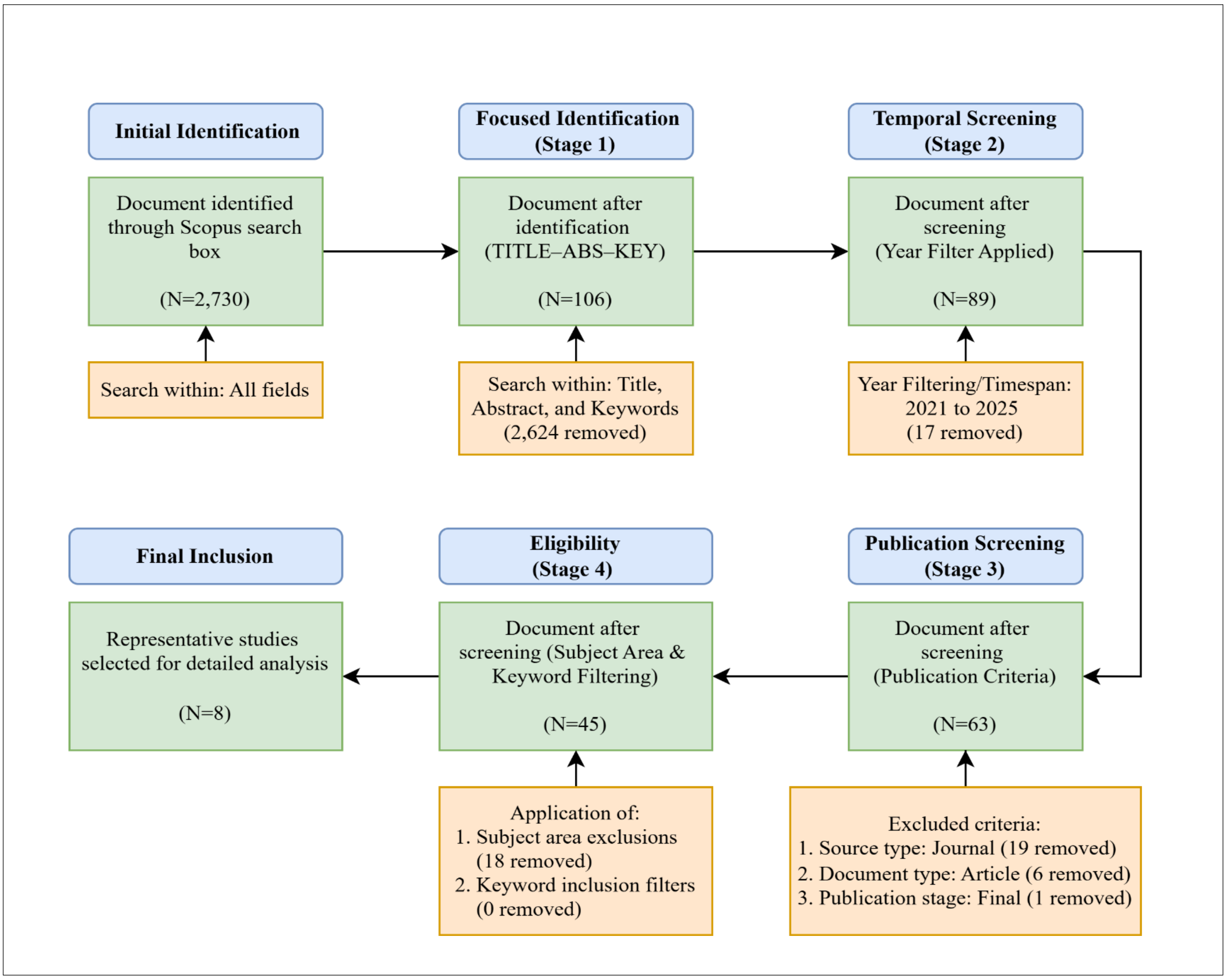

Figure 1 Flowchart of steps followed for the critical study on the use of EEG and ML in diagnosing MDD for the period of 2021 to 2025

analytical tools in the Scopus database. The aim is to examine the publication trends by the distribution of articles by year, country and subject area classification. These tools also provide a visualisation graphs that illustrate the trends.

Each paper also examined critically to find the comparison based on the problem addressed, method employed, EEG signal characteristics and feature extraction techniques, dataset size and validation strategies, and reported performance outcomes. This dual analysis approach enabled both the identification of research trends and a critical assessment of existing solutions, limitations, and unresolved challenges in the field.

III. Result and Discussion

A. *Publication Disribution by Year, Countries, and Subject Area*

From the Scopus database, the annual production of academic documents indexed from 2021 to 2025 related to this study can be tracked in Figure 2.

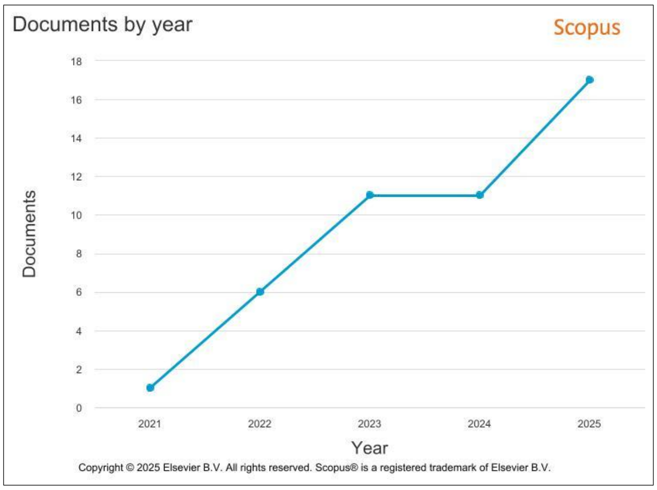

Figure 2 Documents by year

The line graph illustrates a consistent upward trend, beginning with only a single document produced in 2021 and fix with the requirements setup for this critical analysis study. Followed by 6 papers in 2022. Then the volume remained steady with 11 papers between 2023 and 2024. By the end of this five-year span, the total number of published academic papers increase to its peak at 16 papers in 2025. The analysis of publication trends reflects the increase in research and recognition of EEG as a valuable tool for the detecting MDD.

Geographically, most publications from 2021 to 2025 are originated from China, United States and South Korea. Within the five years of period, 20 papers have been contributed from China reflecting strong national investment in artificial intelligence and neuropsychiatric research. Additionally, United States has contribute 7 research papers showed a steady and consistent contribution across the years. South Korea with 5 papers in publications showing the expanding research efforts in biomedical signal analysis and mental health applications. This indicate that Asia and North America are leading in this emerging area. The remaining papers are from the other 7 countries as shown in Figure 3 which are Czech Republic, United Kingdom, Canada, Hong Kong, India, Pakistan, and Australia.

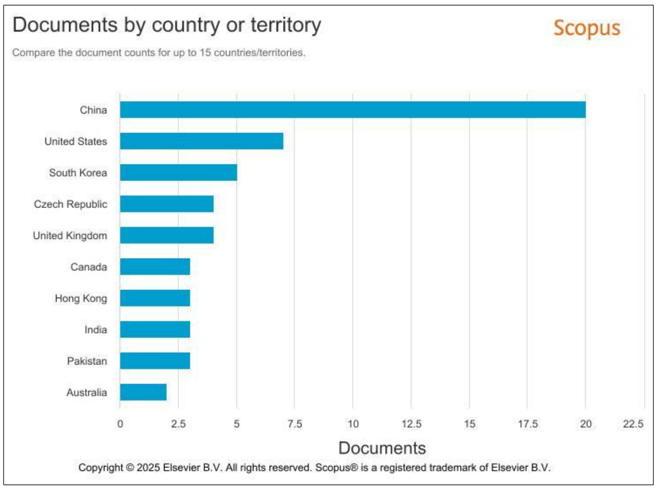

Figure 3 Documents by countries or territory

Figure 4 presents the distribution of publications by subject area for studies on EEG-based machine learning approaches to MDD detection between 2021 and 2025. The figure highlights the interdisciplinary nature of this research field, with contributions spanning medical, engineering, and computational disciplines.

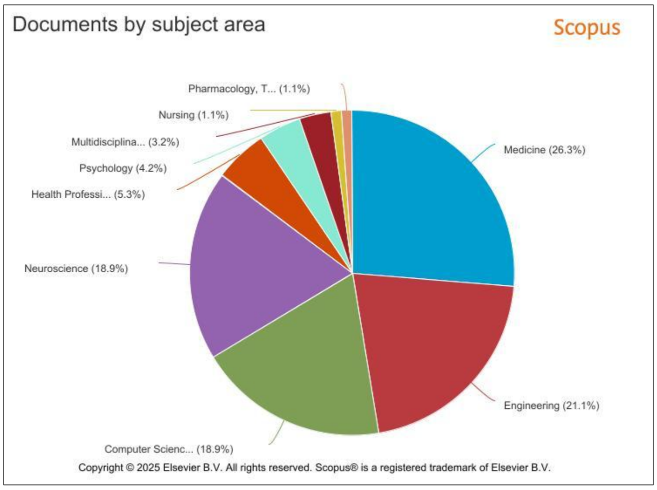

Figure 4 Documents by subject area

Medicine accounts for the largest proportion of publications (26.3%), indicating that EEG-based MDD detection is primarily driven by clinical and diagnostic motivations. This reflects the strong emphasis on improving objective assessment tools for psychiatric disorders and integrating machine learning into clinical decision-making processes. Engineering (21.1%) and Computer Science (18.9%) collectively represent a substantial share of the literature, underscoring the role of algorithm development, signal processing techniques, and model optimization in advancing EEG-based diagnostic frameworks.

Neuroscience also contributes significantly (18.9%), highlighting the importance of understanding brain mechanisms and neural biomarkers underlying MDD. Smaller but notable contributions comes from Psychology (4.2%), Health Professions (5.3%), and Multidisciplinary research (3.2%) further demonstrate the field's cross-domain integration. The relatively lower representation from areas such as Nursing and Pharmacology suggests that EEG-based machine learning for MDD remains predominantly technology and diagnosis-oriented, with future opportunities for broader integration into patient care, treatment monitoring, and clinical practice.

B. Methodological Trends in EEG-Based MDD Detection

In order to identify Major Depressive Disorder (MDD), researchers have employed a variety of techniques. EEG-based machine learning studies for MDD detection published between 2021 and 2025 show a distinct methodological evolution. The sophisticated approach has replaced the traditional machine learning (ML) architectures. Movahed et al. employed RBFSVM, a traditional machine learning technique that combines support vector machines (SVM) with a classifier called RBF kernel [1]. To identify MDD, Li et al. employed machine learning techniques such as Support Vector Machine (SVM), logistic regression (LR), and linear regression (LNR) [6]. To increase the accuracy of depression recognition, Deng et al. employed a convolutional neural network (CNN) as a deep learning (DL) technique called SparNet, which consists of five parallel convolutional filters [4]. Furthermore, in 2023, Aderinwale et al. classified patients with Parkinson's disease (PD) and MDD from healthy people using non-linear EEG features using SVM and the method applied to two-channel EEG data obtained from the frontal lobes of 149 participants [9]. Using single-channel ECG data, Habib et al. introduced MDDBranchNet, a parallel branch deep learning model for MDD binary classification [8]. The R-R interval signal and the degree distribution time series from the horizontal visibility graph are the other two ECG-derived signals incorporated into the model. For the diagnosis of MDD based on EEG signals, Anik et al. proposed an extended 11-layer one-dimensional DCNN (Ex-1DCNN) that enables automatic feature learning without the need for manual feature selection [5]. To accurately diagnose MDD, Naregalkar et al. used state-of-the-art tree-based ensemble models, specifically XGBoost and CatBoost, and derived seven time-frequency and non-linear EEG variables [11]. Wang et al. proposed an M-MDD multi-task deep learning framework [7]. The framework presents a Supervised Feature Extraction Task to extract temporal, spectral and spatial EEG features, which are then combined, and a Contrastive Noise Loudness Task to learn noise invariant representations.

Conventional methods rely mainly on feature-based learning, using manually generated features such as statistics, spectra, wavelets, and nonlinear dynamic features, to distinguish patients from healthy controls [1]. By focusing on end-to-end learning to automatically extract intrinsic brain wave patterns from raw or preprocessed data, modern DL models, such as Ex-1DCNN, avoid the cumbersome manual feature selection process [5]. Traditional 10-fold cross-validation [1], [5], [8], [7], and 5-fold separation have been replaced by more stringent subject-independent methods such as Leave-One-Subject-Out (LOSO) and stratified subject separation [4], [6], [9], [11] to ensure model generalizability.

C. Key Challenges in Existing Studies

A review of selected studies highlights several challenges in EEG-based MDD detection research. According to Movahed et al., the stated high accuracy may be less general because the data set studied has a limited number of samples [1]. Publicly available EEG datasets with more subjects are very lacking. The risk of constant overfitting is maintained by using a moderate sample size of the data set. Although combining multiple sets of features can be beneficial, doing so results in a high-dimensional matrix that confuses interpretation and adds computational burden. High-dimensional data will create very large feature search spaces that are computationally complex to reduce, while deep learning models often lack the interpretability required for medical applications and are viewed as "black boxes." This means that the model can achieve very good predictions but may not be able to communicate how the results were achieved. Deep learning models often lose spatial features between brain regions when using raw or frequency domain data. On the other hand, manual feature extraction is criticized for being time-consuming and laborious, and small data sizes force researchers to simplify deep models to avoid training failures [4].

Aderinwale et al. reported that accurate diagnosis of MDD is complicated by high comorbidity with other psychiatric conditions, such as panic disorder [9]. EEG signals suffer from limited signal-to-noise ratio and relatively poor spatial resolution, while small sample sizes limit researchers' ability to draw robust conclusions. Deep learning models also face the risk of overfitting and poor generalization when trained on limited samples in this research. Anik et al. observed that questionnaire-based assessments produce inconsistent results due to the lack of objective biological markers [5]. Hand-crafted features in traditional machine learning are often tedious and unreliable, and many existing deep learning studies rely on modest datasets of only 30 to 64 subjects. Most previous frameworks fail to consider all brain waves holistically, particularly ignoring Gamma rhythms. Habib et al. noted that multi-channel EEG is often not suitable for home-based monitoring due to the difficulty of signal acquisition [8]. Generating manual heart rate variability features requires significant domain knowledge and computational power. Scale-based clinical assessments are subjective and hampered by societal stigma, and there is a severe lack of public data for deep learning research in this area. Naregalkar et al. reported that patient responses during clinical interviews can be misleading or influenced by the subjective judgments of psychologists [11]. Most research is limited to single feature domains or limited brain regions, while certain deep learning architectures, such as ShallowNet, show poor generalizability across datasets. Before advanced models can be used as ready-to-use diagnostic systems, they still need to undergo extensive clinical validation. According to Wang et al., biological artifacts from various sources inevitably contaminate EEG recordings, and it is still difficult to extract and properly fuse temporal, spectral, and spatial features [7]. The functional relationships and significance of various brain regions are ignored when treating channels separately, and supervised models are especially prone to overfitting as a result of noise and sparse data.

A synthesis of eight studies shows that data-related limitations are the most pervasive challenges in EEG-based MDD detection research. Most studies rely on small personal data sets, limiting the generalizability of models and increasing the risk of overfitting, while the lack of large and publicly available EEG data sets prevents cross-validation of studies. Methodological challenges include the use of high-dimensional or hand-crafted features, which increase computational complexity and reduce interpretability, and frequently use deep learning models as black boxes, which limit transparency and clinical acceptance. EEG recordings are also affected by noise and artefacts, and inconsistencies in channel selection, feature extraction and frequency band coverage further complicate model development. Clinical challenges include the subjectivity of questionnaire-based diagnosis, the presence of comorbid conditions, and difficulties in home-based or large-scale monitoring. Overall, these limitations emphasize the need for larger and standardized data sets, robust preprocessing and feature extraction strategies, and clinically interpretable machine learning frameworks to enable real-world applications.

D. Method to Overcome EEG-MDD Challenges

Through the eight studies, researchers implemented a range of solutions to address challenges in EEG-based MDD detection. Movahed et al. tackled the subjectivity of traditional questionnaire-based diagnosis and the inconsistency of single-domain features by developing an integrated framework combining statistical, spectral, wavelet, functional connectivity, and nonlinear analyses [1]. They optimized feature selection with Sequential Backward Feature Selection (SBFS) and applied 1-minute signal slicing for data augmentation to compensate for small sample sizes. Li et al. addressed the lack of medical interpretability and computational complexity of high-dimensional deep learning features by deriving highly MDD-correlated β/α ratio features using linear models such as SVM and LR to enhance transparency [6]. Deng et al. proposed SparNet, a CNN with parallel filters to simultaneously learn local and global space-frequency features, augmented with attention mechanisms to weight important channels and phase-space reconstruction for denoising, thereby overcoming the loss of spatial features and laborious manual extraction [4]. Other studies also focused on simplifying data acquisition and enhancing robustness. Aderinwale et al. implemented a 2-channel frontal EEG setup that suitable for primary care and applied a wavelet-chaos approach with LASSO regression for robust feature selection to account for comorbid conditions [9]. Anik et al. developed an 11-layer Ex-1DCNN for automated end-to-end learning from raw brainwaves, holistically analyzing all frequency bands and optimizing epoch duration to improve accuracy, particularly highlighting the Gamma rhythm [5]. Habib et al. addressed the challenge of multi-channel EEG unsuitability for home monitoring by proposing MDDBranchNet, a parallel-branch deep learning model using single-channel ECG augmented with R-R signals and Horizontal Visibility Graph degree distribution [8]. Naregalkar et al. applied tree-structured ensemble models such as XGBoost and CatBoost, extracting features across time, frequency, and nonlinear domains to overcome the restriction to single-feature spaces and improve interpretability [11]. Finally, Wang et al. developed M-MDD, a multi-task framework integrating a Contrastive Noise Robustness Task and a CNN-Transformer hybrid encoder to fuse temporal, spectral, and spatial features, addressing the challenges of biological noise and complex feature fusion [7].

Collectively, these solutions reflect a practical and multifaceted effort to address common challenges such as limited data, lack of interpretability, and signal noise. Approaches including signal slicing, single channel acquisition, clinically meaningful indices, attentional mechanisms and multi-task learning show a clear shift from manually designed features towards more flexible deep learning frameworks. These developments allow the model to handle EEG data more reliably and improve the consistency and accuracy of depression detection.

IV. Conclusion

This review analysed EEG-based machine learning studies for Major Depressive Disorder published between 2021 and 2025, highlighting publication trends, methodological approaches, and key research challenges. The results show a clear upward trend in research activity, particularly after 2022, confirming that EEG-based ML for MDD detection is an emerging interdisciplinary field involving medicine, neuroscience, computer science, and engineering. Methodologically, the literature shows a transition from traditional machine learning models using handcrafted features toward deep learning architectures capable of automated feature extraction and multi-domain fusion. These approaches have proved promising classification performance, suggesting the potential of EEG as an objective biomarker for MDD when combined with advanced computational techniques.

However, despite methodological progress, several critical limitations exist. Most studies still rely on small and private datasets, leading to overfitting and limited generalisability, while EEG signal noise and the absence of standardised preprocessing pipelines hinder reproducibility. In addition, the lack of model interpretability and insufficient clinical validation restrict real-world applicability. Future research should focus on building large, and standardised public EEG datasets, developing explainable and clinically transparent models, and conducting model validation studies so the EEG-based model could be used in the clinical settings practically. Using techniques such as transfer learning, federated learning, and multi-task frameworks may further address data scarcity and privacy constraints, further accelerate the translation of EEG-based machine learning systems into practical clinical tools for MDD diagnosis.

Overall, this analysis highlights the importance of data quality, interpretability, and validation in real-world settings, emphasizing that clinical adoption necessitates more than just technological performance. One crucial lesson is that the successful translation of EEG-based ML systems requires close collaboration between engineers, doctors, and healthcare organizations. Future systems, machine, or devices should be created as clinical decision-support tools integrated into existing diagnostic workflows rather than as stand-alone prediction models in order to encourage early screening and monitoring of MDD in hospitals and community mental health settings.

Acknowledgment

The author would like to express his sincere appreciation to the lecturer of Special Topics in Computer Science (CSC649) for the guidance and support throughout this study. Gratitude is also extended to all individuals who contributed directly or indirectly to the completion of this work.

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